

How Long Do I Want to Fade Away? The Duration of Fade-To-Black Transitions in Target-Based Discontinuous Travel (Teleportation)

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ABSTRACT

A fade-to-black animation enhances the transition during teleportation, yet its duration has not been systematically explored even though it is one of the central parameters. To fill this gap, we conducted a small study to determine a preferred duration. We find a short duration of 0.3 s to be the average preference, contrasting durations used previously in the literature. This research contributes to the systematic parameterization of discontinuous travel.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual Reality; Human-centered computing—Interaction techniques; Human-centered computing—Empirical studies in HCI;

1 INTRODUCTION

Teleportation is one of the most common methods of locomotion in VR applications. It is easy to use and not cybersickness-inducing. The method consists of two main phases, which are a target specification (selection) and transition [9]. While the change of the user's position happens instantaneously and is the namesake for the methods, i.e., usually without any continuous movement, neither the target specification nor the transition process themselves need to be discrete. A common method of realizing the transition phase is a simple fade in and out to black, between the change of position (cf. [4]). A fade-to-black animation is often used to visualize a teleportation transition providing the user with feedback that a movement has taken place and at the same time soften the otherwise abrupt change of viewpoints [2]. Although the fade-to-black transition is widely used, there is no systematic investigation into its parameterization. An inevitable parameter to its implementation is the duration of the fade animation. The lack of research into this parameter motivates us to tackle the following research question:

RQ: Which duration is most preferred for a fade-to-black transition when using Point & Teleport? With this investigation, we aim to contribute to a systematic parameterization of discontinuous locomotion methods using a fade-to-black animation.

2 RELATED WORK

To the best of our knowledge, there is no prior work explicitly investigating a preferred duration for a fade-to-black animation in discrete virtual locomotion techniques. There are, however, works that integrate this animation into their locomotion method and mention its parameters. We therefore want to take a look at different implementations of fade-to-black animations used throughout literature.

While a fade-to-black animation can be described by a multitude of parameters, e.g., its color, the duration is one of the key variables. It is usually a constant time in which the users field of view fades linearly from a clear field of view into a fully blocked view and back.

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Figure 1: The virtual environment used in the study.

Lindal et al. [6] use a constant delay with a total duration of 1.6 s. Drogemuller et al. use 1.0 s [3] as a duration. Rahimi et al. use 1.5 s [7] and Freitag et al. use a fade but do not specify the exact duration [5].

3 METHOD

To evaluate a preferred duration for the fade-to-black animation, we conducted a study. Participants are allowed to teleport freely on a sparsely equipped path environment using a Point & Teleport interface extended with a fade-to-black transition. To select a preferred duration for the fade animation, participants may increase or decrease the duration of the fade animation in steps of 0.1 s until a preferred duration is found.

3.1 Procedure & Task

In the beginning, participants are asked to indicate their gender, age, and experience level with VR. Participants are then immersed in a virtual environment that consists of a straight path and a few trees to the side of the path (see Figure 1). To move in this environment a simple Point & Teleport locomotion method with a fade-to-black animation during the transition phase is used. The target specification of this teleportation is solved by a parabolic ray with a maximum range of 11 m, which is a common choice for this parameter (c.f. [1]).

Participants are asked to move freely and adjust the duration of the animation. The duration is initially set to 0.4 s as this was perceived best in an internal test phase. The 0.4 s consist of a fade-out and a fade-in taking up half the total duration each. The total duration can be adjusted in steps of ± 0.1 s with a minimum duration of 0.1 s. Once a preferred duration is set, the application is terminated, and the study ends.

3.2 Participants

We tried to keep the sample diverse within the possibility of a limited sample size, considering factors such as age, experience with VR, and gender. This led to a total of 9 participants (5 male, 4 female) between the ages of 25 and 67 years ($M = 39.5$ years, $SD = 15.91$ years) being recruited for the study. All participants were reached through the personal and professional networks of the experimenter. A normal or corrected vision was a prerequisite for participation in the study. There was no compensation for participation. Of the nine participants,

Table 1: Results for each participant with the participant's age, gender, experience level with virtual reality systems in range [1,3] (1 $\hat{=}$ no experience, 2 $\hat{=}$ occasional usage and 3 $\hat{=}$ heavy usage), the preferred total duration for the fade animation and weather no fade is preferred initially.

#	Age	Gender	Experience	Preferred duration	No fade
1	36	m	3	0.7 s	
2	37	f	2	0.1 s	y
3	25	m	3	0.3 s	
4	60	f	1	0.5 s	
5	67	m	1	0.1 s	
6	27	f	3	0.3 s	y
7	30	m	3	0.3 s	
8	25	f	3	0.3 s	y
9	39	m	3	0.2 s	

six reported having a lot of experience with virtual reality, one person reported occasional use, and two people reported no experience.

4 RESULTS

In response to question **RQ**, the preferred duration for the fade-to-black animation, the following results are aggregated: $M = 0.31$ s, $SD = 0.13$ s, $Md = 0.3$ s, $Min = 0.1$ s, $Max = 0.7$ s. A complete description of every participant's response can be found in Table 1. In additional comments, three participants (participants #2, #6, and #8) express a preference for a duration of 0 s, i.e., no visible fade at all, if it was possible. For these participants, the preferred duration in Table 1 describes their preference if a fade is inevitable. Participant #1 finds a too-fast animation bothersome during frequent teleportation, while a long waiting time is also perceived as bothersome. Participant #3 finds a too-long animation uncomfortable because it triggers blinking.

5 DISCUSSION

Our results show a preference for a total duration of $M = 0.31$ s, $SD = 0.13$ s. The median $Md = 0.3$ s aligns with the aggregated mean value and the relatively small standard deviation. We therefore argue that 0.3 s is the preferred duration for a fade-to-black animation, answering our main research question **RQ**.

Looking at different implementations of a fade-to-black animation in the literature, we see a disparity between our results and the used parameterization. The longest duration found throughout the literature is used by Lindal et al. [6] with 1.6 s while the shortest duration is that of Drogemuller et al. [3] with 1 s. One could also consider an instantaneous transition as a fade-to-black animation with a duration of 0 s. A multitude of implementations use this approach. However, since this does not help in visualizing the transition, we do not further consider this.

The nonzero values are higher by more than a factor of three compared to our results. The longest single preference we record is 0.7 s which is still lower than all of the durations used in other works. Our results, therefore, indicate that the duration used in previous works might be generally too high. Bozgeyikli et al. [2] notice impatience in participants who have to wait for a fade to complete. This observation indicates as well that the duration of an animation could be shortened. Similarly, in our results, we find three participants preferring a duration of 0 s, i.e., no fade at all, and one participant remarking that a long duration feels bothersome.

To evaluate our results different limitations have to be discussed. To establish a preferred duration for a fade-to-black animation, we deliberately excluded 0 s to ensure a meaningful visualization of the transition and a softening of the otherwise abrupt viewpoint change. This goal is not achieved with a duration of 0 s. Despite the restriction, one-third of participants preferred no fade, questioning if the prohibition is justified. Interestingly, two-thirds of those choosing no fade alternatively chose 0.3s over the minimum 0.1s, suggesting

that, if a fade is needed, a 0.3 s duration might be suitable.

A limitation that might affect our results is the virtual environment used in this study. The sparsely equipped environment possibly increased the need for animation during the transition as no animation at all can lead to the transition being unnoticeable. The environment does not allow for orientation based on special landmarks that would hint at the relative displacement of a user after teleportation.

An aspect that is not covered by our study is the reasoning behind the choice of a preferred duration. As Rieke and Zielasko [8] mention, the choice of a locomotion method is heavily dependent on the use case. Since a fade-to-black animation can be part of a teleportation interface, it is not far-fetched to think that its usefulness and duration depend on the scenario as well. Further research should therefore clarify which scenarios benefit most from a fade-to-black animation and if a preferred duration changes dependently.

6 CONCLUSION

With this work, we contribute to the systematic parameterization of fade-to-black animations. Through a study conducted, we answer our research question for a preferred duration of a fade-to-black animation with a duration of 0.3 s. Previous research employed much longer durations.

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REFERENCES

- [1] J. Bhandari, P. MacNeilage, and E. Folmer. Teleportation without Spatial Disorientation Using Optical Flow Cues. In *Proc. of Graphics Interface*, pp. 162 – 167. Canadian Human-Computer Communications Society, 2018. doi: 10.20380/GI2018.22
- [2] E. Bozgeyikli, A. Raij, S. Katkooi, and R. Dubey. Point & Teleport Locomotion Technique for Virtual Reality. In *Proc. of ACM Symposium on Computer-Human Interaction in Play*, pp. 205–216. ACM, 2016. doi: 10.1145/2967934.2968105
- [3] A. Drogemuller, A. Cunningham, J. Walsh, B. H. Thomas, M. Cordeil, and W. Ross. Examining Virtual Reality Navigation Techniques for 3D Network Visualisations. *Journal of Computer Languages*, 56:100937, 2020. doi: 10.1016/j.col.2019.100937
- [4] N. Feld, P. Bimberg, B. Weyers, and D. Zielasko. Keep it simple? Evaluation of Transitions in Virtual Reality. *Proc. of ACM CHI Extended Abstracts*, 2023. doi: 10.1145/3544549.3585811
- [5] S. Freitag, B. Weyers, and T. W. Kuhlén. Interactive Exploration Assistance for Immersive Virtual Environments Based on Object Visibility and Viewpoint Quality. In *Proc. of IEEE Conference on Virtual Reality and 3D User Interfaces*, pp. 355–362, 2018. doi: 10.1109/VR.2018.8447553
- [6] P. J. Lindal, K. R. Jóhannsdóttir, U. Kristjánsson, N. Lensing, A. Stühmeier, A. Wohlan, and H. H. Vilhjálmsson. Comparison of Teleportation and Fixed Track Driving in VR. In *Proc. of IEEE International Conference on Virtual Worlds and Games for Serious Applications*, pp. 1–7. IEEE, 2018. doi: 10.1109/vs-games.2018.8493414
- [7] K. Rahimi, C. Banigan, and E. D. Ragan. Scene Transitions and Teleportation in Virtual Reality and the Implications for Spatial Awareness and Sickness. *IEEE Transactions on Visualization and Computer Graphics*, 26(6):2273–2287, 2020. doi: 10.1109/TVCG.2018.2884468
- [8] B. E. Rieke and D. Zielasko. Continuous vs. Discontinuous (Teleport) Locomotion in VR: How Implications can Provide both Benefits and Disadvantages. In *Proc. of IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops*, pp. 373–374. IEEE, 2021. doi: 10.1109/vrw52623.2021.00075
- [9] T. Weissker, A. Kunert, B. Frohlich, and A. Kulik. Spatial Updating and Simulator Sickness During Steering and Jumping in Immersive Virtual Environments. In *Proc. of IEEE Conference on Virtual Reality and 3D User Interfaces*, pp. 97–104. IEEE, 2018. doi: 10.1109/vr.2018.8446620